



[Practice Challenge](#) [Session 1 Recap](#)

Programming Fundamentals

Selection 1 — Decisions with if and if-else

Programming Fundamentals Team

SETU

2026-09-09



Outline

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23. Fix the Program

**24. Mini-program:
Ticket Price**

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Outline

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3. Flow of Control

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8. = versus ==

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16. When We Need Two Outcomes

17. if-else

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1. Today

By the end of this session you should be able to:

- Explain sequence and selection.
- Recognise Boolean conditions.
- Use relational operators correctly.
- Write `if` and `if-else` statements.
- Trace a program and predict which branch runs.
- Use indentation correctly in Python.



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2. Warm-up: What happens next?

```
name = input("Name: ")
print("Hello", name)
print("Welcome to SETU")
```



2. Warm-up: What happens next?

```
name = input("Name: ")  
print("Hello", name)  
print("Welcome to SETU")
```

Every statement runs, from top to bottom.

This is sequence: the default flow of execution.



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3. Flow of Control

Sequence	Statements run in order
Selection	Statements run only when a condition is satisfied
Iteration	Statements are repeated



3. Flow of Control

Sequence	Statements run in order
Selection	Statements run only when a condition is satisfied
Iteration	Statements are repeated

Today we move from programs that always do the same thing to programs that can **make decisions**.



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4. A Real Decision

A cinema might ask:

```
age = int(input("Enter your age: "))
```

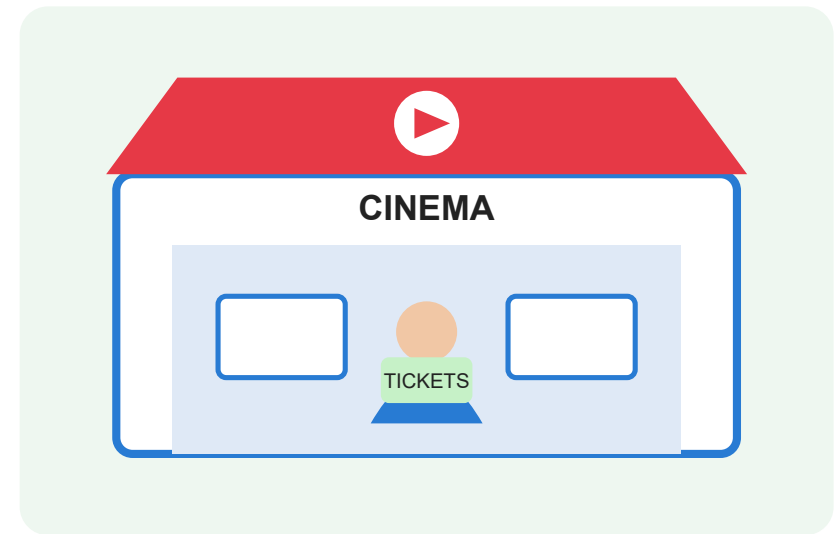
4. A Real Decision

A cinema might ask:

```
age = int(input("Enter your age: "))
```

- If age is 18 or more → allow entry.
- Otherwise → refuse entry.

The program needs a condition whose answer is **True** or **False**.





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5. Boolean Values

Python has a Boolean data type: `bool`.

```
is_raining = True
```

```
logged_in = False
```



5. Boolean Values

Python has a Boolean data type: `bool`.

```
is_raining = True
```

```
logged_in = False
```

There are only two Boolean values:

- `True`
- `False`



5. Boolean Values

Python has a Boolean data type: `bool`.

```
is_raining = True
```

```
logged_in = False
```

There are only two Boolean values:

- `True`
- `False`

Notice the capital letters.



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6. Boolean Expressions

A Boolean expression produces True or False.

```
age >= 18
```

```
mark >= 40
```

```
password == "python"
```

```
number < 0
```



6. Boolean Expressions

A Boolean expression produces True or False.

```
age >= 18
```

```
mark >= 40
```

```
password == "python"
```

```
number < 0
```

Try these in the Python REPL:

```
>>> 10 > 5
```

```
True
```

```
>>> 10 == 5
```

```
False
```



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7. Relational Operators

Operator	Example	Meaning
>	<code>x > y</code>	greater than
>=	<code>x >= y</code>	greater than or equal to
<	<code>x < y</code>	less than
<=	<code>x <= y</code>	less than or equal to
==	<code>x == y</code>	equal to
!=	<code>x != y</code>	not equal to



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8. = versus ==

age = 18

means **put 18 into age.**



8. = versus ==

age = 18

means **put 18 into age.**

age == 18

asks **is age equal to 18?**



8. = versus ==

age = 18

means **put 18 into age**.

age == 18

asks **is age equal to 18?**

Common beginner error: = assigns; == compares.



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9. Predict the Result

What does each expression produce?

`10 > 3`

`7 == 7`

`5 != 5`

`12 <= 10`

`"cat" == "cat"`



9. Predict the Result

What does each expression produce?

`10 > 3`

`7 == 7`

`5 != 5`

`12 <= 10`

`"cat" == "cat"`

Answers: True, True, False, False, True.



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10. The if Statement

```
if condition:  
    statement
```



10. The if Statement

```
if condition:  
    statement
```

Example:

```
age = 20
```

```
if age >= 18:  
    print("You are an adult")
```



10. The if Statement

```
if condition:  
    statement
```

Example:

```
age = 20
```

```
if age >= 18:  
    print("You are an adult")
```

The indented statement runs **only if** the condition is True.



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11. Indentation Matters

Correct:

```
if mark >= 40:  
    print("Pass")  
    print("Well done")
```



11. Indentation Matters

Correct:

```
if mark >= 40:
    print("Pass")
    print("Well done")
```

Incorrect:

```
if mark >= 40:
print("Pass")
```

NOTE:

- Python uses indentation to define the block belonging to the `if`.
- Indentation is a whitespace at the beginning of the line.



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12. Trace an if

```
temperature = 25
```

```
if temperature > 20:  
    print("Warm day")
```

```
print("Finished")
```



12. Trace an if

```
temperature = 25
```

```
if temperature > 20:  
    print("Warm day")
```

```
print("Finished")
```

Output:

Warm day
Finished



12. Trace an if

```
temperature = 25
```

```
if temperature > 20:  
    print("Warm day")
```

```
print("Finished")
```

Output:

Warm day

Finished

What changes if temperature = 12?



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13. One-way Selection

An if statement is **one-way selection**.

```
balance = 100  
withdrawal = 40
```

```
if withdrawal <= balance:  
    balance = balance - withdrawal
```

```
print(balance)
```




13. One-way Selection

An if statement is **one-way selection**.

```
balance = 100  
withdrawal = 40
```

```
if withdrawal <= balance:  
    balance = balance - withdrawal
```

```
print(balance)
```

If the condition is false, Python simply skips the indented block.



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14. Mini Activity: Free Delivery

An online shop gives free delivery when an order is €50 or more.

Complete:

```
order_total = float(input("Order total:  
€"))  
# your selection here
```



14. Mini Activity: Free Delivery

An online shop gives free delivery when an order is €50 or more.

Complete:

```
order_total = float(input("Order total:
€"))
# your selection here
```

Write the condition first. Then decide what belongs inside the block.





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15. Activity Solution

```
order_total = float(input("Order total: €"))

if order_total >= 50:
    print("Free delivery")
```




15. Activity Solution

```
order_total = float(input("Order total: €"))
```

```
if order_total >= 50:  
    print("Free delivery")
```

Question: what happens for an order of €30?



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16. When We Need Two Outcomes

Suppose every mark must be classified as either:

- Pass
- Fail



16. When We Need Two Outcomes

Suppose every mark must be classified as either:

- Pass
- Fail

We could write two separate if statements...

```
if mark >= 40:  
    print("Pass")  
if mark < 40:  
    print("Fail")
```

But Python gives us a clearer structure.



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17. if-else

```
if condition:
    statements_when_true
else:
    statements_when_false
```




17. if-else

```
if condition:
    statements_when_true
else:
    statements_when_false
```

Exactly **one** branch runs.



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18. Pass or Fail

```
mark = int(input("Enter mark: "))

if mark >= 40:
    print("Pass")
else:
    print("Fail")
```



18. Pass or Fail

```
mark = int(input("Enter mark: "))
```

```
if mark >= 40:  
    print("Pass")  
else:  
    print("Fail")
```

Trace it for:

- mark = 72
- mark = 38
- mark = 40



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19. Boundary Values Matter

Consider:

```
if age > 18:  
    print("Adult")  
else:  
    print("Not adult")
```




19. Boundary Values Matter

Consider:

```
if age > 18:  
    print("Adult")  
else:  
    print("Not adult")
```

What happens when `age == 18`?



19. Boundary Values Matter

Consider:

```
if age > 18:  
    print("Adult")  
else:  
    print("Not adult")
```

What happens when `age == 18`?

If 18 counts as adult, the condition should be:

```
if age >= 18:
```

Always test values **at the boundary**.



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20. Strings in Conditions

Selection is not limited to numbers.

```
answer = input("Continue? y/n: ")
```

```
if answer == "y":  
    print("Continuing...")  
else:  
    print("Stopping...")
```



20. Strings in Conditions

Selection is not limited to numbers.

```
answer = input("Continue? y/n: ")
```

```
if answer == "y":  
    print("Continuing...")  
else:  
    print("Stopping...")
```

String comparisons are case-sensitive: "Y" != "y".



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21. Making String Input Friendlier

```
answer = input("Continue? y/n: ").lower()

if answer == "y":
    print("Continuing...")
else:
    print("Stopping...")
```



21. Making String Input Friendlier

```
answer = input("Continue? y/n: ").lower()

if answer == "y":
    print("Continuing...")
else:
    print("Stopping...")
```

Now Y and y are both treated as "y".



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Outline

22. Common Errors

23. Fix the Program

24. Mini-program: Ticket Price

25. Think Before You Code

26. Practice Challenge

27. Session 1 Recap



22. Common Errors

Which errors can you spot?

```
age = input("Age: ")
```

```
if age = 18
    print("You are 18")
else
    print("You are not 18")
```



22. Common Errors

Which errors can you spot?

```
age = input("Age: ")
```

```
if age = 18
    print("You are 18")
else
    print("You are not 18")
```

Problems:

- `input()` returns a string.
- `=` should be `==` for comparison.
- `:` is missing.
- indentation is missing.



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23. Fix the Program

Correct version:

```
age = int(input("Age: "))

if age == 18:
    print("You are 18")
else:
    print("You are not 18")
```



Outline

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24. Mini-program: Ticket Price

Requirement:

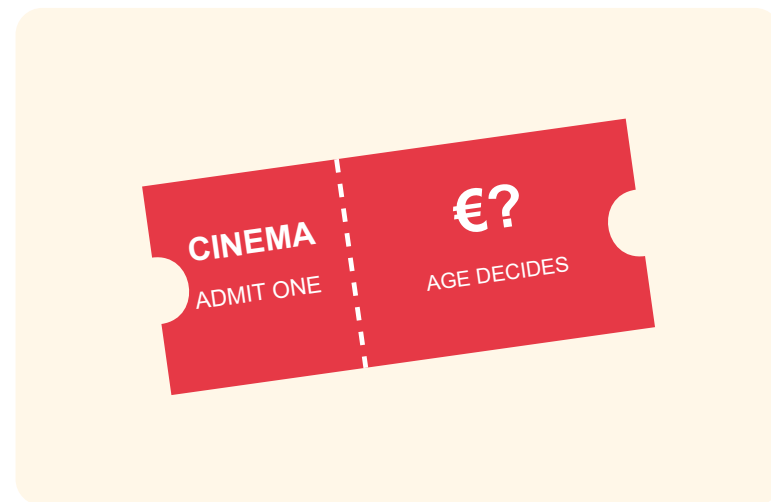
- Under 16 $\rightarrow$ €8
- Everyone else $\rightarrow$ €12

24. Mini-program: Ticket Price

Requirement:

- Under 16 → €8
- Everyone else → €12

```
age = int(input("Age: "))  
if age < 16:  
    price = 8  
else:  
    price = 12  
print("Ticket price: €", price)
```





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25. Think Before You Code

For each problem, identify:

1. What data do I need?
2. What is the Boolean condition?
3. What happens when it is true?
4. Do I need an else?



25. Think Before You Code

For each problem, identify:

1. What data do I need?
2. What is the Boolean condition?
3. What happens when it is true?
4. Do I need an else?

This planning habit becomes increasingly important as selection gets more complex.



Outline

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26. Practice Challenge

Write a program that asks for a number.

- If it is positive, print `Positive`.
- Otherwise print `Zero or negative`.



26. Practice Challenge

Write a program that asks for a number.

- If it is positive, print `Positive`.
- Otherwise print `Zero` or `negative`.

Extension: how could you distinguish between **zero** and **negative**?

That leads us into the next session.



Outline

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27. Session 1 Recap

You should now be comfortable with:

- Boolean values and conditions
- `>`, `<`, `>=`, `<=`, `==`, `!=`
- `if`
- `if-else`
- indentation
- tracing branches
- testing boundary values

Next: **more than two choices** and **combining conditions**.